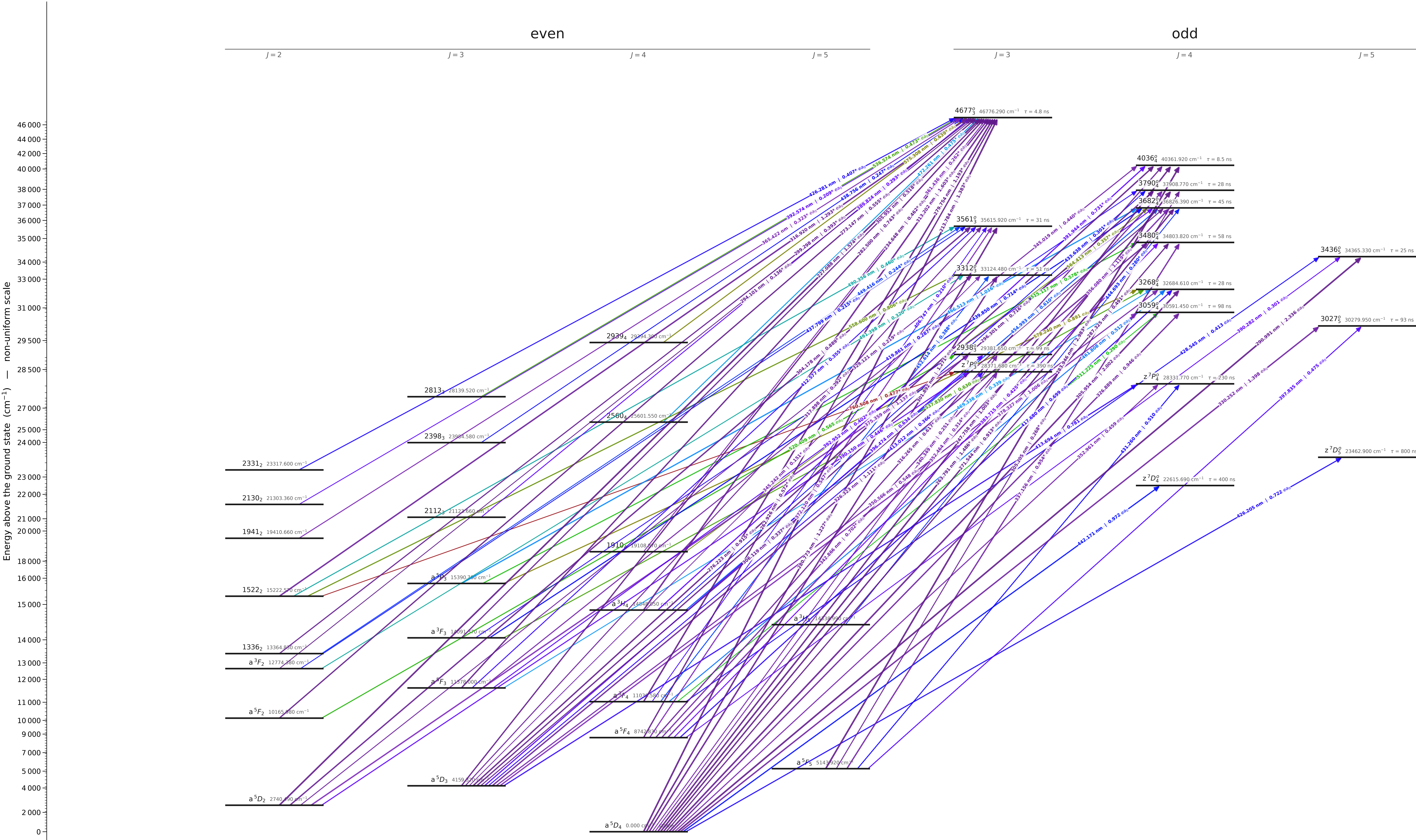
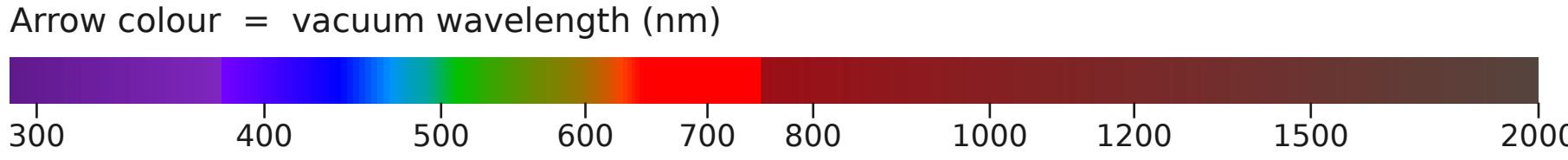


Osmium

Os (Os I). Strongest classified transitions below 2 μm (NIST): 40 levels, 90 lines.

Os⁺ ionisation limit 68058.90 cm⁻¹ = 8.43823 eV (146.932 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[J||e\mathbf{r}||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er J'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁵ D ₄ - 4036 ₄	247.7583	247.6835	1210.0199	1.003*	1.490e+07	18.72	12.7	first-step line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).